SHREYASH SINGHAI

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EDUCATION

- **VIT University** *2023–2027*
  B.Tech in Computer Science & Engineering (AI & ML) | CGPA: 8.24/10
- **Delhi Public School, Bhilai** *Bhilai, India*
  Class XII – 81% (2023) | Class X – 79.8% (2021)

TECHNICAL SKILLS

Languages: Python, SQL, C++

LLM & GenAI: RAG, Agentic AI, Prompt Engineering (few-shot, chain-of-thought, structured outputs), Evaluation (Hit Rate@K, MRR, ROUGE), open-source LLMs (Qwen via Hugging Face)

Frameworks/Libraries: LangChain, Hugging Face Transformers, Scikit-learn, Pandas, NumPy

Databases/Cloud: MySQL, Oracle Cloud Infrastructure

Tools: Git, Google Colab, Google AI Studio, Antigravity, n8n

PROJECTS

- **Sentinel-Code** [GitHub] *Sep 2026 – Present*
  - Built a token-efficient AI security auditor using LangChain, Python, and Gemini 2.5 Flash to automatically detect security vulnerabilities and anti-patterns in Git diffs against custom security rules.
  - Designed a language-aware local pre-filtering engine that filters out trivial changes (formatting, non-code files, and clean comments) locally in under 1 ms per commit, achieving a 48% estimated token reduction across test histories.
  - Integrated zero-config Git pre-commit hooks that intercept commits locally, performing file-aware context structuring and rule-based audit reporting to block high-risk flaws prior to pushing.
- **Terminal-RAG: CLI Retrieval-Augmented Generation Engine** [GitHub] *Jul 2025 – Aug 2026*
  - Architected a local command-line RAG application in Python using Hugging Face Transformers and Qwen 2.5 (0.5B) Instruct, enabling fully offline document querying directly from the terminal.
  - Constructed a vector retrieval pipeline using SentenceTransformers (all-MiniLM-L6-v2) and Scikit-Learn cosine similarity, utilizing line-by-line chunking to maximize context relevance and extraction precision.
  - Streamlined local embedding indexing with a timestamp-based pickle caching mechanism, eliminating redundant vector recalculations and optimizing startup initialization speed over local text databases.
  - Evaluated system accuracy through a diagnostic benchmark harness measuring retrieval and generation metrics, analyzing Hit Rate@K, MRR, Precision/Recall, Separation Gap, and ROUGE scores.
- **Item Recommendation Engine using KNN** [GitHub] *Apr 2025 – May 2025*
  - Implemented a personalized item recommendation engine employing unsupervised K-Nearest Neighbors (NearestNeighbors with Euclidean distance) to suggest products based on user profile similarity.
  - Formulated data preprocessing workflows using Pandas and Scikit-Learn's LabelEncoder to cleanse customer profiles, segment age values into cohorts, and encode multi-class categorical features.
  - Integrated a dynamic inference interface combining live user profile details and real-time seasonal variables (computed via system datetime) to fetch the top 5 closest neighbor items.

ACHIEVEMENTS

- **Pre-Final Round Qualifier, Bain Case Competition:** Qualified for the pre-final round of the national-level case study challenge hosted by Bain & Company.

EXTRACURRICULAR ACTIVITIES

- **Team Lead – Social Media & Analytics**, MERN Stack Club, VIT University *Apr 2025 – May 2026*
  - Collaborated cross-functionally within a technical club to plan and execute data-driven campaigns, translating stakeholder goals into measurable KPIs and driving an 18% increase in event participation.

CERTIFICATIONS

- **Oracle Generative AI Professional** (Oracle) *2025*
  Certified in LLM architectures, RAG, prompt engineering, fine-tuning, and LangChain integration on OCI.
- **Oracle Agentic AI Professional** (Oracle) *2026*
  Certified in multi-agent architectures, ReAct reasoning, tool orchestration, enterprise RAG, and OCI AI governance.